



LAS-SCR: knowledge-driven enhanced legal judgment prediction via the semantic and similar case retrieval of law articles

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Abstract

Legal judgment prediction (LJP) is a fundamental task in legal artificial intelligence. LJP is used to predict relevant law articles, charges, and terms of penalty on the basis of the factual description of legal cases. Existing approaches fail to make full use of external legal knowledge, and rarely pay attention to judges who refer to historical similar cases before they make decisions in real judicial scenarios. To simulate this process, this study proposes a dual-stage framework that integrates law article semantics and similar case retrieval (LAS-SCR). This framework generates initial predictions by deeply fusing case facts with law article semantics through cross-modal attention and hierarchical feature augmentation and, targeting easily confused charges identified from the initial output, it corrects these predictions using contrastive learning-based similar case retrieval. Experiments on two real-world legal datasets demonstrate that our method achieves better results than several baseline models do. Furthermore, our method is effective at distinguishing easily confused charges.

Keywords Legal judgment prediction · Law article semantic fusion · Similar case retrieval · Knowledge-driven

1 Introduction

As societal demands evolve, courts are handling a growing number of cases. To ease the burden on legal professionals and increase efficiency, the legal system is turning to AI. Legal judgment prediction (LJP) and similar case retrieval (SCR) constitute two central research directions. As illustrated in Fig. 1, LJP is used to predict judicial outcomes for given cases, including charges, law articles, and terms of penalty (Xiao et al. 2018). SCR retrieves precedent cases that

exhibit similar factual patterns, legal reasoning, and judicial decisions.

LJP has been extensively applied in the Legal AI domain. Early research relied on rule-based or statistical methods to extract patterns from cases and make predictions (Nagel 1960; Ulmer 1963). Subsequently, machine-learning algorithms enhanced the modeling of complex data relationships (Liu and Hsieh 2006; Martin Katz 2013; Jin et al. 2024). The emergence of deep learning has enabled transformers and pretrained language models (PLMs) to capture semantic information embedded in legal text (Luo et al. 2017; Shen et al. 2018; Yue et al. 2021; Almuzaini and Azmi 2023; Wang et al. 2025; Yahya 2025). Recently, large language models (LLMs), prompt learning, and agents have become the dominant research focus (Peng and Chen 2024; Huang et al. 2025; Song et al. 2025; Yuan et al. 2026).

However, these approaches have two limitations. First, they do not fully leverage the semantic information of legal articles, limiting their understanding of the role of the law in judgment. Second, they neglect the common judicial practice of consulting historical similar cases, which prevents models from utilizing valuable information in precedents.

In real-world judicial practice, judges do not issue immediate verdicts for complex cases. They consult historical precedents to inform their decisions. This observation motivates our innovation: a predict-then-correct paradigm that

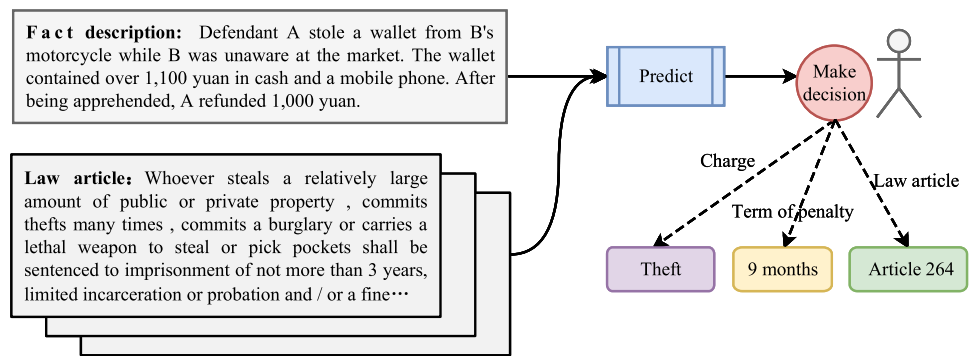
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Fig. 1 An example of legal judgment prediction



mirrors expert judicial reasoning. This paradigm fundamentally diverges from conventional retrieval-augmented methods (e.g., CR4LJP) that inject external knowledge at the input stage. Instead, LAS-SCR operates at the decision level. It first generates an initial prediction and then selectively refines it using targeted case retrieval. This approach marks a conceptual shift from static knowledge augmentation to dynamic, outcome-level correction. Such a design more closely emulates the iterative reasoning of human judicial experts.

Specifically, we develop LAS-SCR, a judicially-inspired framework that combines statutory understanding with precedent-based reasoning. The first stage establishes semantic connections between factual descriptions and law articles through cross-modal interaction, producing preliminary outcomes via task-specific feature decoupling. The second stage dynamically activates a precedent-guided refinement process, where identified confused charges undergo correction through contrastive retrieval of historical cases.

Our work makes the following contributions. (1) We propose LAS-SCR, a dual-stage framework that integrates law article semantics with similar case retrieval. (2) We introduce a novel confusion-aware correction mechanism. Unlike conventional methods that integrate retrieved information at the input, our mechanism utilizes retrieval results at the outcome level to refine predictions. (3) The results of experiments on the CAIL2018 dataset demonstrate the effectiveness of our model.

The remainder of this paper is organized as follows: Section 2 reviews the related literature on LJP and SCR. Section 3 details the proposed methodology. Section 4 presents the experimental setup and results. Section 5 concludes with insights and future directions for LJP.

2 Related work

2.1 Legal judgment prediction

LJPs are critical components of intelligent judicial systems. Their methodologies progress from early rule-based or

statistical approaches, to machine learning and deep learning, and then to methods based on LLMs. Ulmer (1963) uses discriminant analysis to predict the votes of individual justices on the basis of case facts. This work provides early quantitative evidence that judicial outcomes follow systematic patterns. However, these early approaches show limited robustness to data noise and often struggle with complex cases. Later, researchers introduced machine learning methods to LJP tasks. For example, Martin Katz (2013) uses a random forest (RF) model on patent docket data to predict case outcomes and duration. A main limitation of these machine learning models is their reliance on shallow textual features, which restricts their ability to process nuanced legal texts.

The rapid development of deep learning has led to significant advances in LJP performance. Such approaches can be categorized into text-oriented streams and graph-oriented streams. Luo et al. (2017) use attention mechanisms to capture the associations between case facts and relevant law articles for charge prediction. Similarly, Shen et al. (2018) integrate features from law articles with factual descriptions using an end-to-end memory network. To address the issue of overlooking the criminal context, Yue et al. (2021) propose a circumstance-aware framework to mitigate long-text noise and confusion among similar charges. With respect to graph-oriented methods, Wang et al. (2025) develop a framework that uses graph reasoning and supervised contrastive learning to model the dependencies among different subtasks. Dong et al. (2024) link legal cases into a graph and use contrastive learning to cluster related cases. Although these deep learning models effectively capture deep semantic relationships, their decision-making processes often lack transparency and interpretability.

The emergence of LLMs has created new opportunities for LJP. Sun et al. (2024) propose a series of prompt templates designed for LJP. Huang et al. (2025) refine the prompting strategy by introducing a dual-slot mechanism and an integrated scoring approach. Building upon prompt-based methods, other studies incorporate external knowledge sources to enhance prediction quality. Peng and Chen (2024)

present Athena, a framework that retrieves information from a legal knowledge base in conjunction with LLaMA-2 model. Nigam et al. (2025) design a retrieval-augmented generation (RAG) pipeline for Indian common law, enriching case facts with relevant law articles and precedents to emulate realistic courtroom scenarios. Beyond retrieval, some works emphasize interpretability and reasoning. Li et al. (2025) propose a framework that integrates knowledge from multiple sources to provide explanations behind judicial outcomes. Meng et al. (2025) combine prompt learning with graph representations and law article embeddings.

2.2 Similar case retrieval

Early approaches to SCR predominantly use statistical machine learning methods based on shallow textual features, such as term frequency-inverse document frequency (TF-IDF) and topic modeling. Carneiro et al. (2011) employ term frequency-based Bayesian statistics to measure case similarity, whereas Ashley and Walker (2013) utilize feature-driven nearest neighbor algorithms.

With the advent of deep learning, researchers have explored more sophisticated text representation methods for SCR. Shao et al. (2020) propose the BERT-paragraph-level interactions model (BERT-PLI), which captures semantic relations at the paragraph level to infer overall case relevance. Building on this, Hu et al. (2022) design a fact-aware method that processes long-text inputs by encoding query-candidate paragraph pairs. Gao et al. (2023) employ a graph matching network to compute graph-level similarity and retrieve similar cases. Extending these efforts, Liu et al. (2024) present a self-supervised hybrid-retrieval pipeline that alternates between sparse retrieval and dense reranking for Chinese legal cases. Most recently, the focus has shifted toward LLM-driven and RAG-based methods. Park et al. (2025) apply RAG with LLMs to simulate reliable legal reasoning networks within Korean criminal cases. Similarly, Banerjee et al. (2025) combine a fine-tuned Pegasus model with a RAG in a two-stage summarization-and-retrieval approach for Calcutta High Court verdicts, substantially improving both summary fidelity and case recall.

3 Methodology

In this section, we describe our two-stage framework in detail. We first formalize the task and then present a solution composed of multiple modules. We also explain how to correct predictions for confused cases via historical similar case retrieval. Mathematical formulations are provided for clarity.

3.1 Problem formulation

The task of LJP is defined as follows. Given a target case q that contains a factual description f of the defendant's criminal acts, the goal is to predict three outcomes: the relevant law articles y'_l , the charges y'_c , and the term penalty y'_t . This task is formalized as a prediction function P , which learns from a historical case set s to map input facts to the output predictions:

$$P(f) = (y'_l, y'_c, y'_t) \quad (1)$$

3.2 Method overview

The theoretical novelty of LAS-SCR lies in its confusion-aware correction mechanism. This mechanism diverges from static knowledge integration by dynamically activating retrieval only for semantically confused charges. This approach mitigates error propagation and aligns with the cognitive process of legal experts.

Our framework operates in two stages to predict legal judgments as shown in Fig. 2. The first stage comprises three key components: (1) **law semantic fusion (LSF)**, which establishes semantic alignment between case facts and law articles through cross-modal attention mechanisms; (2) **hierarchical feature augmentation (HFA)**, which generates task-specific representations for predicting law articles, charges, and penalty terms through dedicated processing pathways; and (3) **adaptive task balancing (ATB)**, which dynamically coordinates the learning process across the three subtasks through uncertainty-aware loss weighting. The second stage consists of the **contrastive similar case retrieval (CSCR)**, which identifies easily confused charges from the first stage and corrects them through the retrieval and fusion of relevant historical cases. These components are detailed in the following sections. To further clarify our architectural innovations, Table 1 provides a component-level comparison.

3.3 Initial prediction based on law article semantic fusion

3.3.1 Law semantic fusion (LSF)

The LSF module is designed to achieve deep semantic alignment between case facts and law articles through joint textual encoding and a cross-modal attention strategy. This approach contrasts with conventional methods, which treat law articles superficially. For instance, LADAN (Xu et al. 2020) represents law articles as abstract symbols, and models such as FLA (Luo et al. 2017) rely solely on article identifiers. Our LSF facilitates fine-grained semantic correspondence,

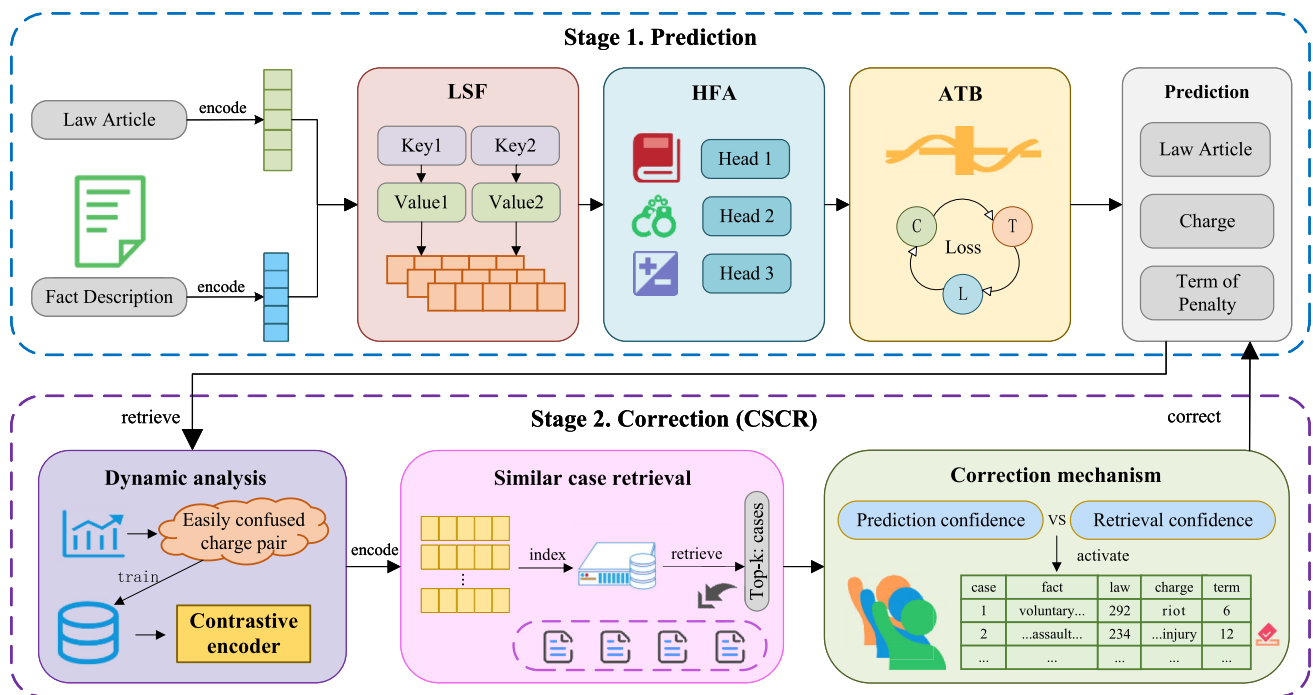


Fig. 2 Two-stage LAS-SCR framework. Stage1 (prediction) and Stage2 (correction). The upper panel illustrates the cross-modal attention mechanism for law-fact alignment, while the lower panel depicts the confusion-aware case retrieval and weighted voting process

enabling the model to pinpoint which specific phrases in a fact description correspond to legal elements within law articles.

Specifically, the LSF module first encodes the case fact description f using a RoBERTa model. To capture both syntactic details and high-level semantic patterns, we extract and fuse features from multiple layers of the model, resulting in a comprehensive representation Z_f . (The detailed formulation is provided in Appendix A.1.) Concurrently, all law articles are pre-encoded offline using a frozen-parameter BERT model to generate their semantic vectors E_a , which enhances computational efficiency.

Next, a cross-modal attention mechanism aligns the fact representation Z_f with the set of law article vectors E_a . Here, Z_f acts as a query to identify the most relevant legal context from the articles. We employ a standard scaled dot-product attention for this alignment. The resulting attention-enhanced context $Attention_{output}$ (Appendix A.3) is then concatenated with the original factual features to form the final representation for the law article prediction, denoted as Z_{law} . This process preserves both fine-grained case details and their legally relevant semantic context.

$$Z_{law} = [Z_f; Attention_{output}] \quad (2)$$

Table 1 Component-Level comparison with prior works

	Novelty	Adaptation	Reuse	Prior Work Comparison
LSF	Cross-modal attention for fact-law article semantic alignment.	Extends standard attention to legal domain by fusing multi-layer RoBERTa features.	RoBERTa encoder.	LADAN : Treats law articles as symbols. FLA : Uses only article IDs.
HFA	Task-specific decoupled pathways for subtasks.	Designs dedicated heads for each subtask with progressive fusion.	GELU, ReLU, LayerNorm.	TopJudge : Shared features across tasks. NeurJudge : Segment-based processing.
ATB	Uncertainty-aware dynamic loss weighting.	Adapts Bayesian principles to multitask legal prediction.	Cross-entropy loss.	MPBFN : Static loss weights. CL4LJP : No task balancing mechanism.
CSCR	Confusion-aware correction with hard-negative suppression.	Integrates contrastive learning with legal confusion patterns.	FAISS for similarity search.	CR4LJP : Injects retrieved cases at input. CL4LJP : General contrastive learning.

3.3.2 Hierarchical feature augmentation (HFA)

In the legal domain, law articles, charges, and penalty terms operate at different semantic granularities. Law articles demand precise matching, charges focus on abstract behavioral patterns, and terms require a holistic synthesis of case context, legal application, and charge severity. A unified feature representation is inadequate for this domain because it causes feature interference and overfitting, ultimately diminishing the model's ability to capture the nuanced legal reasoning required for accurate judgments.

The innovation of HFA lies in its task-specific feature decoupling and progressive fusion architecture. Unlike conventional multitask learning, which shares a single feature space, HFA explicitly constructs separate processing pathways for each subtask, allowing them to specialize in their unique semantic requirements. For law article prediction, a deep mapping network is employed to perform rigorous, high-resolution feature extraction, ensuring precise alignment with statutory text. For charge prediction, HFA utilizes a lightweight feature-selection mechanism. This pathway efficiently filters and refines raw factual features to identify the most discriminative cues related to criminal actions, preventing the model from being overwhelmed by irrelevant details. The term of the penalty prediction head is the most sophisticated component. It first integrates core case features Z_f , the attention-enhanced legal context $Attention_{output}$, and the refined charge features through multi-source feature concatenation. This combined representation is then processed by a network with ReLU activation to capture the complex, non-linear interactions among various sentencing factors. (The detailed network architectures for each head are provided in Appendix A.2.)

By progressively fusing information, from basic facts to the legal context and finally to charge-specific features, HFA emulates the hierarchical reasoning process of a human judge. This design ensures that each subtask receives a tailored representation, enhancing the model's capacity for nuanced legal reasoning.

3.3.3 Adaptive task balancing

In multitask learning, harmonizing the learning process is a significant challenge. This is particularly true for scenarios involving disparate subtasks, such as predicting law articles, charges, and terms of penalty. Different tasks often exhibit varying learning dynamics and convergence rates. If not managed carefully, this can lead to optimization instability. Conventional methods often rely on static, manually tuned loss weights. These methods fail to adapt to evolving gradients, resulting in suboptimal convergence.

To address this issue, we introduce the ATB module, which employs an uncertainty-aware dynamic weighting

mechanism. Instead of fixed weights, ATB treats each task's loss weight as a learnable parameter, represented by a task-specific uncertainty measure. This allows the model to autonomously adjust the influence of each task's loss function on the basis of its current learning state. This mechanism is formalized in the total loss function:

Conventional multi-task learning approaches tend to use fixed weights or empirical manual adjustments. These methods struggle to dynamically adapt to gradient variations across tasks during training and often trap model optimization in local optima. The long-tail distribution of term-of-penalty task may lead to a large initial gradient amplitude. It may lead direction of parameter update and inhibit optimization of law articles and charges. To address this, we propose an uncertainty-aware dynamic weighting mechanism that automatically balances multi-objective optimization through learnable task variance parameters. The adaptive loss function is defined as:

$$\mathcal{L}_{\text{total}} = \sum_{i=1}^3 \frac{1}{2\sigma_i^2} \mathcal{L}_i + \sum_{i=1}^3 \log \sigma_i \quad (3)$$

where σ_i is the uncertainty measure for each task ($i \in \{\text{law, charge, term}\}$), and $\mathcal{L}_i$ is the corresponding cross-entropy loss. This formulation is grounded in the principles of Bayesian deep learning. In this framework, minimizing the negative log-likelihood of the uncertainties acts as a regularizer. This prevents the model from becoming overconfident in any single task.

During backpropagation, the mechanism automatically adjusts task weights. It assigns a smaller weight to the loss of any task exhibiting high uncertainty (large σ_i). This reduces the impact of its noisy gradients, preventing it from destabilizing the shared parameters. Conversely, as a task's optimization stabilizes and its uncertainty decreases (small σ_i), its loss weight increases. This allows the task to contribute more to parameter updates and accelerate convergence. This dynamic balancing act ensures that no single task can dominate the learning process. This is crucial for volatile tasks like terms of penalty prediction. The result is a robust training environment for all legal subtasks. Crucially, the gradients for the uncertainty parameters are calculated exclusively through their respective log terms. This ensures that the core feature learning pathways remain unperturbed by the balancing mechanism itself.

3.4 Contrastive similar case retrieval (CSCR)

In the second stage, we propose a confusion-aware correction mechanism within the CSCR module. This approach emulates the decision-making process of legal experts, who routinely refer to historical precedents to resolve easily

confused charges. The following section introduces the implementation process.

3.4.1 Easily confused pair generation and data construction

Easily confused pair generation In the legal domain, certain charges are inherently confusing because they share core criminal elements but are differentiated by nuanced contextual factors or specific legal definitions. For instance, both “contract fraud” and “fraud” involve the act of deception, being distinguished by whether the act occurs within the context of a contractual relationship. These fine-grained distinctions are often challenging for models to capture from textual descriptions alone.

To identify these challenging distinctions, we employ a bidirectional confusion matrix to map the relationships between predicted and actual charges. This allows us to pinpoint not only which charges are frequently misclassified, but also the specific direction of these errors. To ensure that our findings are robust and not skewed by class imbalance, we calculate a sample-size-weighted error rate for each potential confusion pair. This process dynamically identifies the most problematic charge pairs directly from our model’s prediction behavior on the CAIL-small dataset. The complete list of identified confusing charge pairs and their corresponding error rates is presented in Appendix B, Table 6.

Data construction For each confused charge pair, we construct a contrastive case library. The sampling strategy for this library is informed by the principles of Hard Example Mining (HEM) (Xiong et al. 2022). It includes three distinct types of cases: positive, regular negative, and hard negative. Positive cases are defined as those instances that were correctly classified by the initial model for the actual charge. Regular negative cases consist of examples from other charge categories that are textually dissimilar. Hard negative cases are those that exhibit significant lexical or semantic overlap with the actual charge but are assigned to a different legal category. This structured sampling approach guides the model to differentiate between these semantically proximate cases. This ensures that the correction process is focused on the most challenging scenarios, directly addressing easily confused charges identified in the initial error analysis.

3.4.2 Similar case retrieval

Building upon the easily confused charges, we develop a confusion-aware contrastive learning framework to address the nuanced distinctions between legally similar yet distinct

charges. This approach directly integrates domain-specific knowledge of legal confusion patterns into the training objective. The core of our framework is a modified contrastive loss function to handle the specific challenges of legal charge discrimination. The loss function is formulated as follows:

$$\mathcal{L}_{\text{cont}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\sum_{y_j=y_i} e^{s_{ij}/\tau}}{\sum_{y_j=y_i} e^{s_{ij}/\tau} + \alpha \sum_{y_j \neq y_i} e^{s_{ij}/\tau}} \quad (4)$$

where $s_{ij} = h_i^T h_j$ represents the cosine similarity between case embeddings. To refine this objective for the specific challenges of legal judgment prediction, we introduce two key parameters. τ controls how strictly the model distinguishes between similar and dissimilar cases. By setting this value relatively low, we effectively sharpen the model’s focus, forcing it to make more confident distinctions. This is particularly important for improving the model’s ability to recognize less common, or tail-class, charges, which might otherwise be overlooked. The second parameter is a hard negative suppression coefficient, α . For example, a case of fraud might appear almost identical to a case of contract fraud in its description. A standard contrastive learning approach would heavily penalize the model for considering these cases to be similar, pushing them far apart in the representation space. However, this can be counterproductive, as the model should learn to recognize their subtle legal differences rather than just their textual similarities. The coefficient α mitigates this issue by reducing the penalty for such hard negatives. This allows the model to acknowledge their surface-level similarity while still learning to focus on the critical legal features that differentiate them. Together, these parameters create a more balanced learning process that is better suited to the nuances of legal texts.

During the retrieval phase, we employ a targeted mechanism to support the correction process. Specifically, if the initial prediction from the first stage results in a charge that belongs to one of our pre-identified confused charge pairs, the CSCR module is triggered to retrieve similar cases. This conditional activation ensures that our retrieval resources are focused precisely where the model is most likely to err. The retrieval process proceeds as follows. First, the input case is encoded using the same confusion-aware contrastive encoder, generating a vector representation that captures the legally relevant features for distinguishing similar charges. Second, we leverage the Facebook AI similarity search (FAISS) library to efficiently query a prebuilt index of historical cases, retrieving the top-k most similar prece-

dents. Finally, this initial set of retrieved cases may undergo a refinement process to ensure their relevance and diversity before being used for correction.

3.4.3 Correction mechanism

The CSCR module uses a confidence-based method to decide whether to correct its initial predictions on a case-by-case basis. For each easily confused charge, this mechanism quantitatively compares two distinct sources of confidence: the model's internal prediction confidence and the external consensus confidence derived from retrieved similar cases. Through this case-level dynamic comparison, the system determines on whether to initiate a correction.

The first-stage prediction confidence is directly obtained from the probability distribution of the charges generated by the model for the current case. Specifically, we take the highest probability value p_{max} corresponding to the predicted charge as the base confidence, which is then dynamically adjusted using a bias term λ :

$$C_{pred} = \lambda \cdot P_{max}, \quad (\lambda \geq 1) \quad (5)$$

This bias term λ is not a fixed parameter but is dynamically set according to the historical error frequency of the predicted charge within confusing charge pairs. When $\lambda = 1$, the system applies a standard threshold where correction is triggered whenever the retrieval confidence exceeds the original prediction confidence. When $\lambda > 1$, the retrieval evidence must demonstrate higher confidence to override the initial prediction. This dynamic adjustment ensures stricter correction control for charge pairs with high error rates, effectively preventing the correction mechanism from being misled by the model's own systematic biases. The second-stage retrieval confidence measures the ability of the retrieval module to distinguish between confusing charges through the following formula:

$$C_{ret} = \frac{s_{max}}{s_{max} + s_{second_max}} \quad (6)$$

where s_{max} and s_{second_max} represent the highest and second-highest similarity scores between the input case and cases of different charges in the contrastive case library, respectively. This design ensures that the retrieval confidence genuinely reflects the support strength from historical cases for the current judgment, rather than merely indicating superficial similarity.

The correction mechanism is activated only when $C_{ret} > C_{pred}$. This threshold design follows clear judicial logic: when external case evidence demonstrates higher confidence than the model's internal judgment, it indicates that historical precedents provide more convincing legal grounds.

This case-level triggering mechanism avoids unnecessary corrections while ensuring that each modification is based on sufficient legal evidence.

Once correction is activated, the system generates the final prediction through a weighted voting mechanism. This mechanism systematically integrates the similarity scores of retrieved cases with their corresponding charge distributions. This integration enables the quantitative utilization of historical case information, supporting the final decision in a data-driven manner.

4 Experiments

We conduct comprehensive experiments to evaluate our proposed framework. This section first describes the dataset and the experimental setup. Baseline models for comparative studies are then specified. We also provide an analysis and discussion of the findings.

4.1 Datasets and experimental settings

Datasets This study utilizes the CAIL2018 dataset (Xiao et al. 2018), a public collection of legal cases from the Supreme People's Court of China¹. The dataset comprises two subsets, CAIL-small and CAIL-big, with each case containing factual descriptions, law articles, charges, and penalty terms. We adopt a preprocessing procedure consistent with prior research (Yue et al. 2021; Zhang and Dou 2023; Zhang et al. 2023). We perform the following filtering steps. First, we remove cases with incomplete or inconsistent labels. Next, we exclude cases with multiple law articles or charges to guarantee label uniqueness. We then eliminate law articles and charges that occur in fewer than 100 cases. Finally, the penalty terms are categorized into nonoverlapping intervals. The resulting dataset statistics are presented in Table 2. The CAIL-small subset comprises 95,509 cases covering 87 law articles and 104 charges, whereas the CAIL-big subset is significantly larger, comprising 1,516,995 cases covering 105 law articles and 121 charges. Due to its broad scope and large number of cases, the dataset has become a widely recognized and authoritative resource in legal NLP.

Table 2 Dataset statistics

Dataset	CAIL-small	CAIL-big
Training Set Cases	77168	1359874
Testing Set Cases	18341	157121
Law Articles	87	105
Charges	104	121
Terms of Penalty	11	11

Implementation details For the above approach, we perform training on a 1 vGPU-32GB. We utilize the AdamW optimizer with a learning rate of $2e-5$ and carry out 15 epochs of training. The final model is evaluated on the test dataset.

4.2 Baseline models

We compare our model with several strong baselines:

Topjudge Dong and Niu (2021) presents a topological multitask learning framework that formalizes inter-subtask dependencies via Directed Acyclic Graphs (DAGs) to inject domain priors.

MPBFN Yang et al. (2019) adopts multi-perspective bidirectional feedback networks with word collocation attention to link the subtasks and spot common word patterns.

LADAN Xu et al. (2020) applies graph neural networks to capture discriminative features among confusable law articles.

NeurJudge Yue et al. (2021) implements partitioned fact analysis where each segment contributes to subtask predictions.

CR4LJP Zhang and Dou (2023) integrates case retrieval through top-k similar historical case vectors concatenated with original case descriptions.

CL4LJP Zhang et al. (2023) uses contrastive learning to model relationships between fact descriptions, law articles, and charges.

LKEPL Zhao (2025) enhances a prompt learning model by injecting external legal knowledge for improved prediction performance.

4.3 Evaluation metrics

We employ four widely used multi-classification metrics: accuracy (ACC), macro-precision (MP), macro-recall (MR), and macro-F1 (MF1).

$$\begin{aligned} \text{ACC} &= \frac{\sum_{i=1}^K \text{TP}_i}{N}, \\ \text{MP} &= \frac{1}{K} \sum_{i=1}^K \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i}, \\ \text{MR} &= \frac{1}{K} \sum_{i=1}^K \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i}, \\ \text{MF1} &= \frac{1}{K} \sum_{i=1}^K \frac{2 P_i R_i}{P_i + R_i}. \end{aligned} \quad (7)$$

where TP_i represents the number of samples correctly predicted as class i . FP_i denotes samples from other classes incorrectly classified as class i . FN_i refers to samples of class i that are misclassified.

4.4 Experimental results and analysis

Tables 3 and 4 present the performance comparisons of all the models. Our proposed LAS-SCR model demonstrates comprehensive advantages in both the CAIL-small and the CAIL-big datasets. The following sections provide a detailed analysis of the results for each subtask.

Law article prediction The experimental results show that LAS-SCR outperforms the strong baseline CL4LJP. In a CAIL-small datasets, it achieves gains of 0.04% in accuracy, 0.15% in MP, 2.53% in MR, and 2.28% in MF1. These improvements increase to 1.44%, 1.49%, 5.91%, and 4.33% on the larger CAIL-big dataset. This increase in recall is due to our LSF module. Through a cross-modal attention mechanism, the module jointly encodes the contextual fact

Table 3 Experimental results on CAIL-small datasets (%)

Method	Law Articles				Charges				Terms of Penalty			
	ACC	MP	MR	MF1	ACC	MP	MR	MF1	ACC	MP	MR	MF1
Topjudge	89.40	85.78	83.48	84.30	88.19	85.13	83.31	83.79	36.68	32.96	34.94	32.75
MPBFN	89.44	86.00	84.34	84.78	88.20	85.37	83.31	83.93	36.77	34.17	33.46	33.57
LADAN	81.20	78.24	77.38	76.47	85.07	83.42	82.52	82.74	38.29	36.16	32.49	32.65
NeurJudge	89.92	87.76	86.75	86.96	90.37	87.22	85.82	86.13	41.65	40.44	37.20	37.27
CR4LJP	91.37	88.68	87.85	87.91	89.32	86.75	85.77	85.96	38.02	37.14	33.98	34.31
CL4LJP	91.42	90.31	87.95	88.06	89.90	88.13	85.79	86.80	38.50	38.11	34.16	34.50
LKEPL	89.23	86.47	83.59	84.86	90.12	87.81	86.34	86.75	40.21	39.17	35.63	36.81
Ours	91.46*	90.46*	90.48*	90.34*	90.79*	90.58*	89.19	89.66*	45.03	38.75	36.04	36.69

The best results in each column are shown in **bold**. * denotes statistically significant improvement over all baselines ($p < 0.05$) according to a paired t-test

Table 4 Experimental results on CAIL-large datasets (%)

Method	Law Articles				Charges				Terms of Penalty			
	ACC	MP	MR	MF1	ACC	MP	MR	MF1	ACC	MP	MR	MF1
Topjudge	95.02	86.48	80.21	82.46	94.61	86.43	79.43	82.01	55.74	45.83	40.40	42.06
MPBFN	95.07	87.33	80.54	82.91	94.57	86.56	79.57	81.89	55.83	44.29	41.10	41.21
LADAN	96.57	86.22	80.78	82.36	96.45	88.51	83.73	85.35	59.66	51.78	45.34	46.93
NeurJudge	95.57	85.57	78.81	80.54	95.58	82.01	77.05	78.05	57.07	47.65	40.01	41.18
CR4LJP	95.94	88.50	84.49	85.76	95.24	88.00	82.35	84.36	58.01	48.64	45.37	45.60
CL4LJP	96.33	89.91	84.56	86.39	95.87	88.71	83.15	85.42	58.00	48.57	45.44	45.79
LKEPL	96.51	85.74	77.23	80.39	96.62	88.05	79.07	81.64	58.90	48.64	43.21	44.07
Ours	97.77*	91.40*	90.47*	90.72*	97.67*	92.33*	89.26*	90.54*	62.33*	58.82	56.39	57.31

The best results in each column are shown in **bold**. * denotes statistically significant improvement over all baselines ($p < 0.05$) according to a paired t-test

descriptions and the textual content of law articles. This enables the model to precisely align specific phrases in the fact description with corresponding legal elements in the articles. It helps the model distinguish between articles with overlapping language but different legal definitions. While the MPBFN focuses on local lexical dependencies through word collocation attention mechanisms, it neglects global statistical patterns across law articles. Similarly, the graph networks of LADAN treat law articles as symbols, overlooking the essential semantic details that our joint encoding model successfully captures. These findings demonstrate that our explicit semantic alignment is a more effective strategy for law article prediction.

Charge prediction LAS-SCR improves the MF1 score by 2.86% over CL4LJP in CAIL-small datasets and by 5.12% in the CAIL-big dataset. This gain stems from two main factors. First, our dynamic attention mechanism excels at modeling long-range dependencies within case facts. This capability is crucial for understanding complex legal narratives. In these narratives, a defendant's initial act may be described in one paragraph. However, the consequential outcome or intent might be detailed much later. This mechanism ensures a holistic understanding of the event. In contrast, baseline models like MPBFN, which focus on local word collocations, fail to establish these distant connections. Our model's higher MR provides evidence for this advantage, demonstrating its ability to identify all relevant information scattered throughout the text. Second, the CSCR module offers a targeted correction mechanism. It refines initial predictions for confusing charges by leveraging evidence from similar cases. CR4LJP concatenates retrieved vectors, which risks error propagation. In contrast, our postprediction approach avoids these cascading failures. This design results in more robust charge discrimination.

Term of penalty prediction As the most complex subtask, the penalty prediction term shows the greatest gains from our framework. In the CAIL-big dataset, our model achieves an MF1 of 57.31% and an accuracy of 62.33%. This represents a 10.93% improvement in MF1 over NeurJudge. This performance is attributed to the HFA network. Unlike baselines that mix information, HFA explicitly decouples features from case facts, law articles, and charges. It then progressively integrates them. The module first processes core case facts, then enriches them with legal context, and finally incorporates charge-specific features. This progressive fusion mimics the hierarchical reasoning of a human judge. This design captures the complex, nonlinear relationships required for the terms of penalty. In contrast, the baseline models struggle with this complexity. NeurJudge processes factual segments separately, which can overlook critical cross-segment dependencies. Similarly, CR4LJP concatenates retrieved case features, making it vulnerable to noise from irrelevant information. Our HFA architecture overcomes these limitations, leading to improved performance.

4.5 Ablation studies

To better demonstrate the contribution of individual components in LAS-SCR, we conduct ablation studies in the CAIL-small dataset. We assess the impact on three subtasks by sequentially removing key modules from the model. All the experiments are performed under identical training settings and evaluation metrics. The results are summarized in Table 5.

Removing the CSCR module leads to a slight decrease of 0.24% in the overall Macro-F1 score for charge predic-

Table 5 Ablation study on CAIL-small datasets (%)

Module	Law Articles				Charges				Terms of Penalty			
	ACC	MP	MR	MF1	ACC	MP	MR	MF1	ACC	MP	MR	MF1
LAS-SCR	91.46	90.46	90.48	90.34	90.79	90.58	89.19	89.66	45.03	38.75	36.04	36.69
w/o CSCR	91.17	90.05	90.41	90.06	90.46	90.35	89.03	89.42	45.10	38.74	36.07	36.75
w/o LSF	91.02	89.26	90.17	89.59	90.34	89.68	88.85	89.06	44.47	38.06	34.05	35.38
w/o ATB	90.94	89.96	90.24	89.93	90.40	90.28	88.72	89.39	43.93	37.39	35.06	35.18
w/o HFA	90.97	89.72	90.43	89.97	88.82	86.61	86.71	86.36	44.22	34.43	32.88	30.33

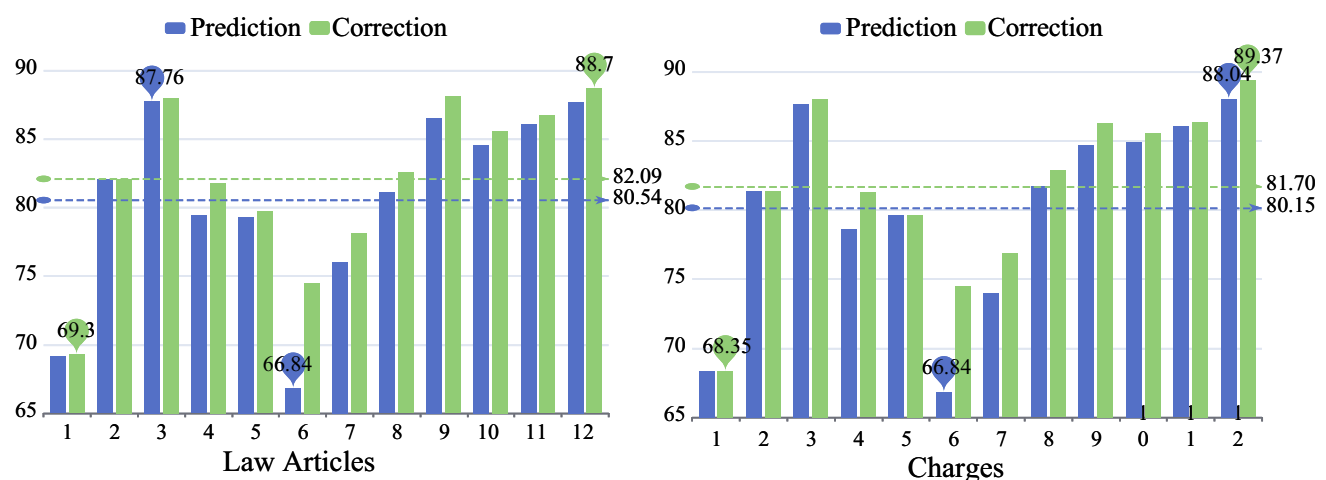
tion. This outcome is expected, as the module is designed for targeted intervention rather than uniform performance enhancement. Its effectiveness stems from a knowledge-driven approach: when an initial prediction falls into a pre-identified easily confused charges, CSCR retrieves semantically similar historical precedents. This process introduces external legal evidence, highlighting subtle legal distinctions that are often indiscernible from fact descriptions alone. By outputting specific historically similar precedents, the model provides transparent, case-based reasoning for the final decision, thereby enhancing its interpretability. Therefore, while its impact on the overall score is modest, the CSCR module improves accuracy for these scenarios. To fully appreciate its contribution, we present a detailed, targeted analysis of its performance on these critical cases in Section 4.6.

Excluding the LSF module results in a 0.75% decrease in law article prediction Macro-F1 because of the lack of explicit fact and law alignment. By leveraging cross-modal attention, the LSF enables the model to focus on legally meaningful textual cues, thereby improving its ability to distinguish between law articles with overlapping elements.

The removal of the ATB mechanism leads to a 1.17% decline in term prediction accuracy. ATB dynamically adjusts the task-specific loss weight based on the current learning

state. If a task exhibits high uncertainty, a common occurrence in complex sentencing scenarios, its loss weight is automatically reduced. This suppression of noisy gradients prevents the penalty prediction task from destabilizing the shared network parameters. As the model's performance on the task improves and its uncertainty diminishes, the loss weight is progressively increased. This allows the task to contribute more effectively to the training process. Ultimately, ATB ensures a balanced optimization by preventing any single task from dominating the gradient updates.

In contrast, eliminating the HFA module causes the greatest performance decrease. The macro-F1 score decreases by 6.42%, demonstrating that the module plays a crucial role in capturing task-specific semantic granularities. Unlike methods that use a single shared feature space, HFA constructs dedicated processing pathways for each subtask. This design prevents feature interference. For law article prediction, HFA employs a deep network that performs high-resolution feature extraction. This ensures precise alignment with the fine-grained text of legal statutes. For charge prediction, it uses a lightweight feature-selection mechanism. This efficiently filters raw facts to identify the most critical behavioral cues, avoiding noise from less relevant information. The most sophisticated design is for terms of penalty prediction. Here,

**Fig. 3** Performance of the CSCR module on easily confused charges. (a) Accuracy for similar law articles. (b) Accuracy for easily confused charges

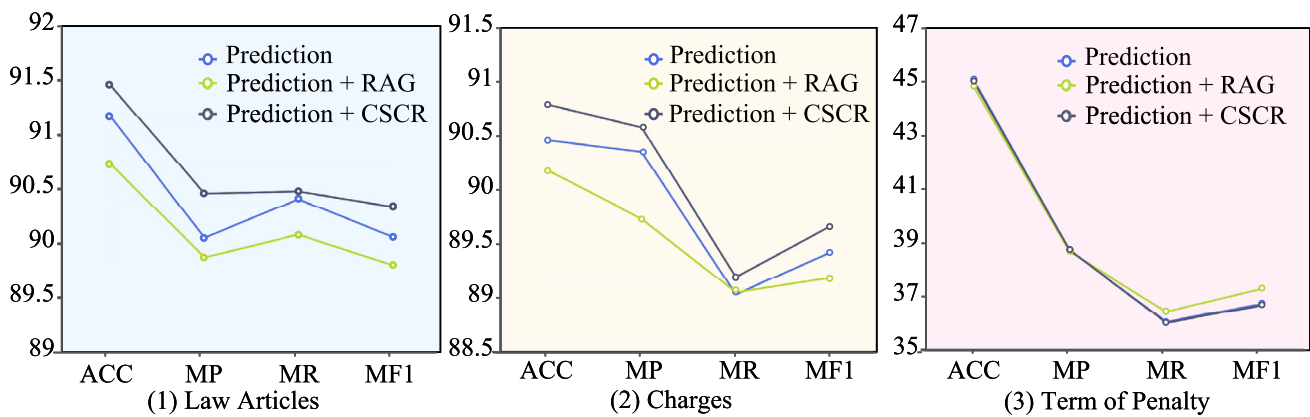


Fig. 4 Comparison of the performance of CSCR and RAG

HFA performs a progressive fusion of multiple sources: case facts, law articles, and charge features. A ReLU-activated network then processes this fused representation to capture the complex, nonlinear interactions among various sentencing factors.

Collectively, these results demonstrate that each component contributes in a distinct yet complementary manner: CSCR enhances fine-grained charge discrimination, LSF reinforces fact and law alignment, ATB stabilizes multi-task optimization, and HFA enables deep legal reasoning. Together, they improve the robustness and interpretability of the overall framework.

4.6 Targeted analysis on easily confused charges

To evaluate the effectiveness of our CSCR module, we conducted a focused analysis on the 12 most error-prone charge pairs identified from our initial model's predictions (as listed in Appendix B, Table 6). For each of these pairs, we compared the prediction accuracy before and after applying the CSCR correction mechanism. The results are shown in Fig. 3. By

retrieving relevant historical precedents, the CSCR module introduces external legal knowledge that guides the model's decision-making. This process improves the performance.

4.7 Comparative study with RAG

To objectively evaluate the CSCR, we compare its performance with that of the end-to-end RAG approach. The experimental results are shown in Fig. 4. CSCR performs better in law article and charge prediction tasks, whereas RAG is advantageous in terms of penalty prediction. This performance difference stems from the distinct knowledge integration mechanisms of the two methods and the inherent characteristics of each task.

CSCR first generates initial predictions on the basis of the fact description and then corrects easily confused charges using the retrieval results. This structure maintains the stable encoding of legal texts by the pretrained model. In contrast, the RAG introduces external knowledge at the input stage. While this aims to enrich contextual understanding, it may disrupt the model's encoding process of original case facts,

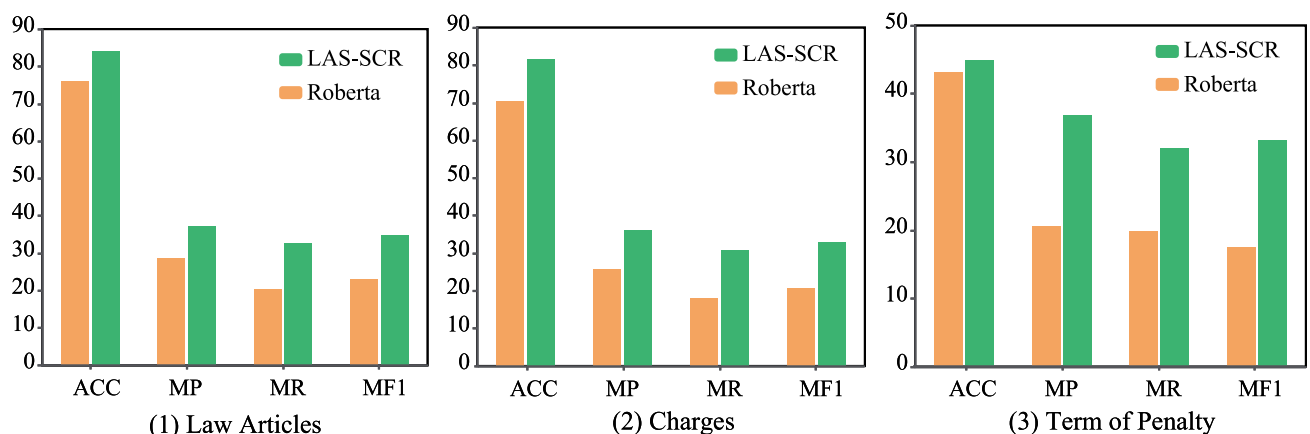


Fig. 5 Performance comparison on rare charges

thereby affecting its ability to capture textual distinctions. In terms of the penalty prediction task, the current approaches generally simplify it into a classification problem with 11 nonoverlapping intervals, without fully accounting for the complex influence of sentencing circumstances. RAG introduces external knowledge that provides valuable references for this task, which relies more on comprehensive judgment, thus contributing to its performance improvement.

This study revealed that the adaptive relationship between knowledge introduction methods and task characteristics affects the performance of the model. The structure of CSCR is more suitable for processing discrimination tasks that require precise semantic alignment, whereas the RAG mechanism has more potential in decision-making tasks that require multiparty information.

4.8 Case study on rare charges

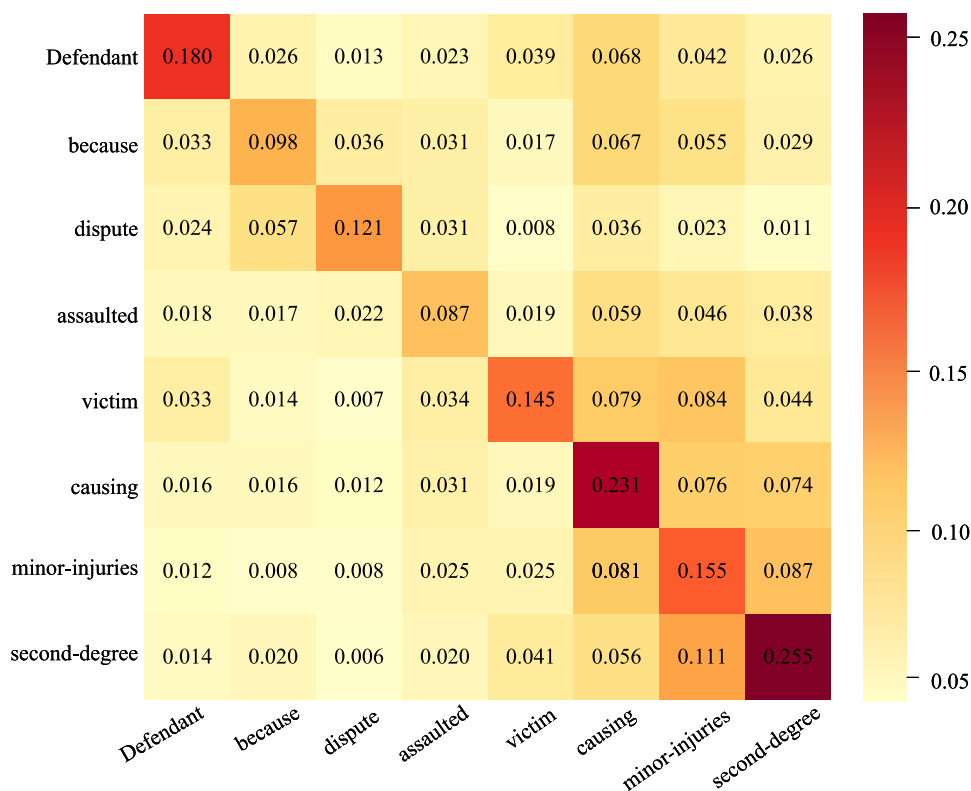
To evaluate the model performance in low-resource scenarios, we selected 10 rare charges with fewer than 100 training samples from the CAIL-small dataset. The proposed LAS-SCR framework was compared against a RoBERTa baseline. As shown in Fig. 5, LAS-SCR consistently outperforms the baseline for these rare charges. These results demonstrate that our method effectively learns discriminative features from limited data, indicating generalization in data-scarce environments.

4.9 Interpretability analysis

We examine the interpretability of the model through token-level attention visualization, as illustrated in Fig. 6, for an intentional injury case. The attention heatmap reveals that the model consistently focuses on legally determinant phrases: “second degree” receives the strongest attention (0.255), aligning with injury severity as defined in Article 234 of the Chinese Criminal Law. Significant attention is also given to “causes” (0.231) and “minor injuries” (0.155), reflecting the emphasis on the causal linkage between acts and consequences. Contextual elements such as “dispute” (0.121) and “assaulted” (0.087) are assigned moderate weights, which is consistent with their role in establishing the circumstances of the crime. The visualization demonstrates how the model processes input text through the distribution of attention weights. This reflects a pattern consistent with the reasoning process in LJP.

To further enhance the interpretability of our model, Fig. 7 illustrates the case-level correction process of the CSCR module. As depicted, the model first predicts the charge of intentional homicide based on the case description, and then triggers the CSCR module. CSCR then computes easily confused charge pairs and retrieves similar historical cases from the case base. When the retrieval confidence c_{ret} is higher than the predicted confidence c_{pred} , the system will ultimately correct the prediction. The calculations for this

Fig. 6 Token-level attention visualization for an intentional injury case



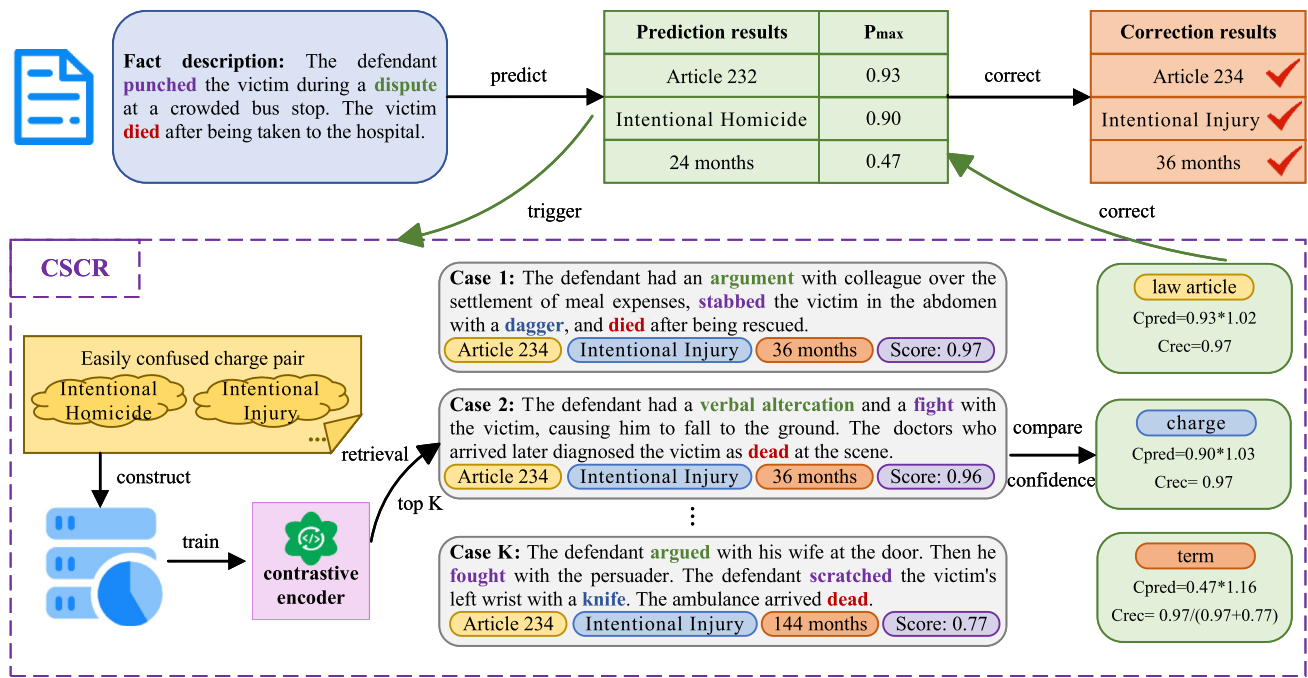


Fig. 7 Case-level correction process of the CSCR module

mechanism are derived from Eqs. (5) and (6), 1.02 is the λ calculated dynamically.

5 Conclusion

In this paper, LAS-SCR, a novel dual-stage framework for legal judgment prediction, is presented. In the first stage, our framework applies semantic fusion techniques to deeply integrate case facts with relevant law articles. This process establishes meaningful connections between factual descriptions and legal provisions, enabling the model to generate accurate initial predictions through cross-modal understanding. The second stage implements a correction mechanism that targets dynamically identified easily confused charges from the first stage. For these specific charges, the system retrieves semantically similar historical cases through contrastive learning. The framework then leverages these relevant precedents to adjust and refine the final predictions. The experimental results show that our method outperforms baseline models. The framework is particularly effective at distinguishing between easily confused charges scenarios. Our future work will focus on expanding a more diverse legal corpus. This includes constructing dedicated datasets for multi-defendant and multi-charge cases, as well as developing a fine-grained benchmark for easily confused charges. These resources will enable us to extend the LAS-SCR framework to handle these complex scenarios. Additionally,

we will explore more transparent and interpretable methods to elucidate the legal reasoning process.

Appendix A Mathematical details of the proposed method

This appendix provides the complete mathematical formulations for the modules described in Section 3.3 of the main text. The details are presented here to improve the readability of the main manuscript while ensuring full transparency and reproducibility of our work.

A.1 Law semantic fusion (LSF)

This section details the encoding and attention mechanisms within the LSF module.

A.1.1 Fact and law article encoding

The case fact description f is encoded using a RoBERTa model. To capture both syntactic and semantic information, we extract features from multiple layers and fuse them:

$$\begin{aligned} \mathbf{H}_f^{(t)} &= \text{BERTLayer}_t(f), \quad t \in \{9, 10, 11, 12\}, \\ \mathbf{h}_f^{(t)} &= \mathbf{H}_f^{(t)}[0, :], \\ \mathbf{Z}_f &= \text{MLP}([\mathbf{h}_f^{(9)}; \mathbf{h}_f^{(10)}; \mathbf{h}_f^{(11)}; \mathbf{h}_f^{(12)}]). \end{aligned} \quad (\text{A1})$$

where:

- f is the input factual description.
- $H_f^{(t)}$ is the output tensor from the t – th layer of RoBERTa.
- $h_f^{(t)}$ is the $[CLS]$ token representation from layer t .
- Z_f is the final fused representation of the case fact.
- MLP denotes a multilayer perceptron.

Concurrently, each law article a from the complete set of articles A is preencoded offline using a frozen BERT model:

$$E_a = \text{MeanPool}(\text{BERT}(a)), \quad \forall a \in A \quad (\text{A2})$$

where:

- E_a is the semantic vector for law article a .
- MeanPool denotes the mean pooling operation over the output token embeddings.

A.1.2 Cross-modal attention

A cross-modal attention mechanism aligns the fact representation Z_f with the set of law article vectors E_a . The fact representation serves as the query, while the law article vectors serve as keys and values.

$$\begin{aligned} Q &= Z_f W_Q, \\ K &= E_A W_K, \\ V &= E_A W_V, \end{aligned} \quad (\text{A3})$$

$$\text{Attention}_{\text{output}} = \text{softmax}\left(\frac{QK^T}{\sqrt{256/4}}\right)V$$

where:

- E_A is the matrix formed by stacking all law article vectors E_a .
- W_Q, W_K , and W_V are learnable projection matrices for queries, keys, and values, respectively.
- d_k is the dimensionality of the key vectors (set to 256/4 in our implementation).

A.2 Hierarchical feature augmentation (HFA)

This section details the network architectures for each of the three task-specific prediction heads within the HFA module.

A.2.1 Law article prediction head

A deep network is used for precise feature extraction for law article prediction.

$$\text{Head}_{\text{law}} = \text{Softmax}\left(W_{\text{law}}^{(2)} \text{Dropout}_{0.2}\left(\text{LayerNorm}\left(\text{GELU}\left(W_{\text{law}}^{(1)} Z_{\text{law}} + b_{\text{law}}^{(1)}\right)\right) + b_{\text{law}}^{(2)}\right)\right) \quad (\text{A4})$$

where:

- Z_{law} is the input feature from the LSF module.
- $W_{\text{law}}^{(1)}, b_{\text{law}}^{(1)}$ and $W_{\text{law}}^{(2)}, b_{\text{law}}^{(2)}$ are the learnable weight and bias matrices for the two linear layers.
- GELU and LayerNorm are the Gaussian error linear unit and layer normalization functions, respectively.
- $\text{Dropout}_{0.2}$ indicates a dropout layer with a probability of 0.2.

A.2.2 Charge Prediction Head

A lightweight network is employed for efficient feature selection in charge prediction.

$$\text{Head}_{\text{charge}} = \text{Softmax}\left(W_{\text{charge}}^{(2)} \text{LayerNorm}\left(\text{GELU}\left(W_{\text{charge}}^{(1)} Z_f + b_{\text{charge}}^{(1)}\right)\right) + b_{\text{charge}}^{(2)}\right) \quad (\text{A5})$$

where the symbols are defined analogously to the law article head, with the input being the raw fact representation Z_f .

A.2.3 Terms of the penalty prediction head

This head integrates multiple sources of information through feature fusion before prediction. First, the features from different sources are concatenated:

$$Z_{\text{term}} = [Z_f; Z_{\text{attn}}; Z_{\text{charge}}^{(1)}] \quad (\text{A6})$$

where:

- Z_f is the original fact representation.
- Z_{attn} is the attention-enhanced legal context.
- $Z_{\text{charge}}^{(1)}$ is the output of the first linear layer in the charge prediction head (i.e., $\text{GELU}(W_{\text{charge}}^{(1)} Z_f + b_{\text{charge}}^{(1)})$).

This fused representation is then processed by a network with a ReLU activation function:

$$\text{Head}_{\text{term}} = \text{Softmax} \left(\mathbf{W}_{\text{term}}^{(2)} \text{Dropout}_{0.4} \left(\text{LayerNorm} \left(\text{RELU} \left(\mathbf{W}_{\text{term}}^{(1)} \mathbf{Z}_{\text{term}} + \mathbf{b}_{\text{term}}^{(1)} \right) \right) + \mathbf{b}_{\text{term}}^{(2)} \right) \right) \quad (\text{A7})$$

where the symbols are defined analogously, with *ReLU* used as the activation function to handle complex feature interactions, and a higher dropout rate of 0.4 is applied.

Appendix B Statistics of easily confused charge

Table 6 presents the 12 most easily confused charge pairs identified from our analysis on the CAIL-small dataset. Here, “error-prone charges” refer to criminal charges for which the model exhibits high prediction error rates. “Confused charges” indicate the most frequently misclassified charge categories corresponding to these error-prone charges. The “error rate” the represents percentage of misclassifications where an error-prone charge is mistakenly predicted as a specific confusing charge, relative to the total misclassification instances of that error-prone charge.

Table 6 Easily confused charges identified from the first-stage predictions on the CAIL-small dataset

Index	Error-prone charges	Confused charges	Error rate (%)
1	Contract Fraud	Fraud	38.22
2	Fraud	Contract Fraud	31.82
3	Intentional Injury	Provocation	76.19
4	Illegal Business	Gambling	41.98
5	Provocation	Intentional Injury	28.79
6	Involuntary Manslaughter	Intentional Injury	36.92
7	Intentional Homicide	Intentional Injury	63.49
8	Riot	Intentional Injury	34.92
9	Traffic Offense	Involuntary Manslaughter	57.63
10	Involuntary Manslaughter	Major Responsibility Accident	37.78
11	Embezzlement	Misappropriation of Funds	32.56
12	Misappropriation of Funds	Misappropriation of Public Funds	61.11

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Code availability Code will be made available on reasonable request.

Declarations

Conflicts of Interest The authors declare that they have no conflicts of interest.

Ethics approval and consent to participate Legal judgment prediction involves high sensitivity, so strict privacy protection is essential in research. Current model is in an early exploratory stage. Since legal case texts may contain personal information, this study uses publicly available, anonymized CAIL2018 dataset. Collected from official judicial platforms, all privacy in dataset was anonymized before release, ensuring research compliance with privacy protection standards.

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